

CivicPulse AI

Predictive Civic Infrastructure Intelligence

SIH Project Details – Thanglish Version

1. Project Overview

CivicPulse AI enbadhu Indian urban local bodies-kku help pannura oru explainable, AI-powered civic infrastructure intelligence platform.

Usually, pothole, waterlogging, blocked drain, road damage, water leakage, streetlight failure maadhiri civic problems vandha piragu dhaan citizens complaint pannuvanga. Government departments complaint vandha piragu action eduppanga.

CivicPulse AI indha reactive system-ai predictive system-aa maathum.

Historical complaints, rainfall data, weather forecast, roads, drains, waterbodies, elevation, population density, critical locations, infrastructure history, recurrence patterns, citizen reports, and optional sensor data-ai analyse panni, civic problem enga, eppo, evlo serious-aa varum-nu AI predict pannum.

Adhukku apram system:

- Endha area high-risk-la irukku-nu identify pannum.
- Problem varum time window-ai predict pannum.
- Population and important infrastructure impact-ai estimate pannum.
- Probable root cause-ai identify pannum.
- Endha issue-ku first action edukkanum-nu recommend pannum.
- Limited municipal workers, time, vehicles, equipment, and budget-ai optimise pannum.
- Different rainfall and intervention scenarios-ai simulate pannum.

2. Final Project Direction

CivicPulse Monsoon

CivicPulse Monsoon enbadhu CivicPulse AI-oda focused and practical version.

Idhu ward-level drainage failure and waterlogging risk-ai 24–72 hours munnadi predict pannum.

Adhu mattum illaama:

- Affected population-ai estimate pannum.
- Nearby schools, hospitals, roads, and important facilities-ai identify pannum.
- Road-drain-waterbody relationship graph moolama probable upstream cause-ai find pannum.
- Municipal officers-kku preventive inspection plan generate pannum.
- Available crews and budget basis-la best action sequence-ai recommend pannum.

3. Core Problem

Indian cities-la waterlogging, blocked drains, road damage, and service failures usually complaint vandha piragu dhaan handle pannapadudhu.

Heavy rain already start aana piragu action edukkumbodhu:

- Problem late-aa detect aagum.
- Emergency repair cost adhigamaagum.
- Same location-la repeated problem vandhaalum separate complaints-aa treat pannapadum.
- Complaint pannaadha areas ignore aagalam.
- Visible symptom mattum repair pannapadum; actual upstream cause fix aagamal irukkalam.
- Limited workers wrong order-la allocate aagalam.
- High population impact irukkura location first action receive pannaama pogalam.

Municipal data different departments-la scattered-aa irukkum. Complaint system, ward records, rainfall data, map data, drainage information, road department records, and field inspection details oru single system-la integrated-aa irukkaadhu.

4. Proposed Solution

CivicPulse AI following data sources-ai combine pannum:

- Historical civic complaints and incidents.
- Rainfall observation and weather forecast.
- Roads, drains, waterbodies, and settlement data.
- Elevation and low-lying area indicators.
- Population density and critical facility locations.
- Infrastructure age and maintenance history, available irundhaal.
- Citizen reports, field inspection photos, and optional sensor readings.

System generate pannura main outputs:

1. Predictive civic-risk map.
2. 24-hour, 48-hour, and 72-hour forecast.

3. Risk probability and confidence level.
4. Prediction-ku main reasons.
5. Estimated affected population and facilities.
6. Probable upstream cause.
7. Priority action list.
8. Crew and budget-based work plan.
9. Rainfall and intervention what-if simulation.
10. Municipal operations dashboard.

5. Example Use Case

Oru ward-la old drainage corridor irukku. Adhu dense residential area pakkathula irukku.

System following signals-ai detect pannudhu:

- Same location-la repeated waterlogging complaints.
- Recent rainfall events-kku apram complaint frequency increase aagudhu.
- Next 48 hours-ku heavy rainfall forecast irukku.
- Area low-lying terrain-la irukku.
- Drainage connectivity weak-aa irukku.
- Nearby drain blocked-aa irukkura possibility irukku.
- Population exposure high-aa irukku.
- School and primary healthcare centre nearby irukku.
- Previous repairs problem-ai permanently solve pannala.

CivicPulse AI decision card:

- Infrastructure risk: High.
- Risk window: Next 24–72 hours.
- Estimated exposed population: 8,000+; scenario-based estimate.
- Historical recurrence: High.
- Nearby critical facilities: One school and one healthcare facility.
- Main factors: Rainfall anomaly, repeated incidents, low elevation, drainage proximity, and population exposure.
- Probable cause: Upstream drain blockage or insufficient drainage capacity.
- Priority: Critical.
- Recommended action: Rainfall varradhukku munnadi upstream drain inspection and cleaning.
- Confidence: Separate-aa display pannapadum.
- Safety note: Final action-ku field verification required.

Important-aa, model validated data illaama 87% maadhiri false precision show panna koodaadhu. Reliable historical labels irundhaal dhaan exact probability use panna vendum. Illaatti risk class, confidence, evidence, and data-quality warning display panna vendum.

6. Main Objectives

- Complaints increase aaguradhukku munnadi civic problems predict pannudhal.
- Drainage and waterlogging high-risk locations identify pannudhal.
- Repeated and related incidents detect pannudhal.
- Population and infrastructure impact estimate pannudhal.
- Probable upstream cause identify pannudhal.
- Risk, impact, recurrence, and preventability basis-la priority set pannudhal.
- Limited municipal crews and budget optimise pannudhal.
- Pre-monsoon preparedness support pannudhal.
- Explainable AI recommendations provide pannudhal.
- Complaint volume mattum base pannama risk assess pannudhal.
- Human verification and accountability support pannudhal.

7. Target Users

- Municipal commissioners.
- Urban local bodies.
- Ward engineers.
- Stormwater drainage department.
- Public works department.
- Water supply department.
- Disaster management cells.
- Smart city command centres.
- Field inspection teams.
- Municipal planning and budget officers.

8. Complete System Workflow

Step 1: Data Collection

Government APIs, municipal files, complaint records, geospatial data, citizen reports, field photos, and optional sensors-lendhu data collect pannum.

Step 2: Data Cleaning

Coordinates, timestamps, issue types, duplicates, missing values, department names, and outdated map layers validate pannum.

Step 3: Civic Event Understanding

Text, voice, image, and location data-ai structured event-aa convert pannum.

Each event-la:

- Issue category.
- Location.
- Time.
- Severity indicators.
- Affected asset or area.
- Evidence quality.
- Department ownership.

irukkum.

Step 4: Incident and Recurrence Detection

Same area and similar time-la vandha multiple reports-ai single incident-aa group pannum.

Same location-la repeated-aa problem varudhaa-nu recurrence pattern detect pannum.

Step 5: Risk Prediction

Oru grid cell, road segment, drainage area, or infrastructure zone future-la critical civic issue experience panna probability-ai model calculate pannum.

Step 6: Impact Estimation

Predicted risk-ai population, schools, hospitals, roads, settlements, and important facilities-oda overlay pannum.

Step 7: Root Cause and Dependency Analysis

Drains, roads, waterbodies, low-lying areas, outlets, and nearby infrastructure relationship-ai analyse panni probable upstream cause identify pannum.

Step 8: Priority Calculation

Each predicted issue-kku following factors base-la priority score calculate pannum:

- Failure probability.
- Severity.

- Population exposure.
- Critical facility exposure.
- Recurrence.
- Forecast rainfall.
- Infrastructure vulnerability.
- Preventability.
- Response time.
- Data confidence.

Step 9: Resource Optimisation

Available crews, skills, travel distance, budget, work hours, ownership, and urgency consider panni best work plan recommend pannum.

Step 10: Scenario Simulation

Officer rainfall intensity, crew count, budget, blocked drain, road closure, or preventive action change panni different outcomes compare panna mudiyum.

9. Risk-Priority Logic

Practical risk priority formula:

Risk Priority = Failure Probability × Impact × Recurrence × Preventability × Data Confidence Adjustment

System one single unexplained score mattum show panna koodaadhu.

Separate-aa following values display panna vendum:

- Failure probability.
- Expected impact.
- Recurrence level.
- Preventability.
- Confidence.
- Main evidence.

10. Impact Estimation

Estimated impact calculate panna system following information use pannum:

- Predicted risk area-kulla irukkura population.
- Nearby schools and hospitals.
- Important roads and transport links.

- Informal settlements and vulnerable communities.
- Water and sanitation assets.
- Expected disruption duration.

Output-ai “estimated exposed population” nu display panna vendum. Guaranteed affected population nu claim panna koodaadhu.

11. Key Differentiating Features

1. Pre-Complaint Risk Forecasting

Complaint volume increase aaguradhukku munnadi risk predict pannum.

Rainfall, recurrence, terrain, network connectivity, population exposure, and infrastructure condition combine pannum.

2. Explainable Risk Card

Every high-priority prediction-kku model yen indha result kuduthuchu-nu explain pannum.

Example:

- Rainfall anomaly high.
- Same location-la repeated incidents.
- Area low elevation-la irukku.
- Drain nearby irukku.
- Population exposure high.
- School or hospital impact zone-kulla irukku.

3. Drainage Dependency Graph

Drains, roads, waterbodies, settlements, outlets, and critical facilities-ai graph-aa connect pannum.

Visible downstream problem mattum repair pannaama actual upstream cause investigate panna help pannum.

4. Spatial Risk Propagation

Waterlogging or service disruption connected roads, drains, and low-lying areas moolama eppadi spread aagalam-nu estimate pannum.

5. Recurrence Intelligence

Repair panniyum same issue thirumba varura locations-ai identify pannum.

Temporary symptom and structural problem difference-ai purinjukka help pannum.

6. Complaint Underreporting Correction

Complaint kammi irukku-nu risk kammi nu assume pannaadhu.

Population, accessibility, reporting coverage, and geographic evidence use panni underreported areas-ai consider pannum.

7. Confidence-Aware Decisions

Risk probability mattum illaama confidence and data quality display pannum.

High risk but low confidence irundhaal “inspect first” recommendation kudukkum.

8. Exposure-Aware Prioritisation

Same severity irukkura rendu issues-la school, hospital, dense settlement, or major road affect aagura issue-kku higher priority assign pannum.

9. Crew Allocation Optimisation

Available workers, skills, distance, budget, urgency, and expected risk reduction base-la daily work order generate pannum.

10. Intervention Simulation

Oru drain clean pannina enna impact? Extra crew add pannina enna change? Rainfall adhigamaana enna nadakkum? nu simulate panna mudiyum.

11. Multi-Hazard Relationship Analysis

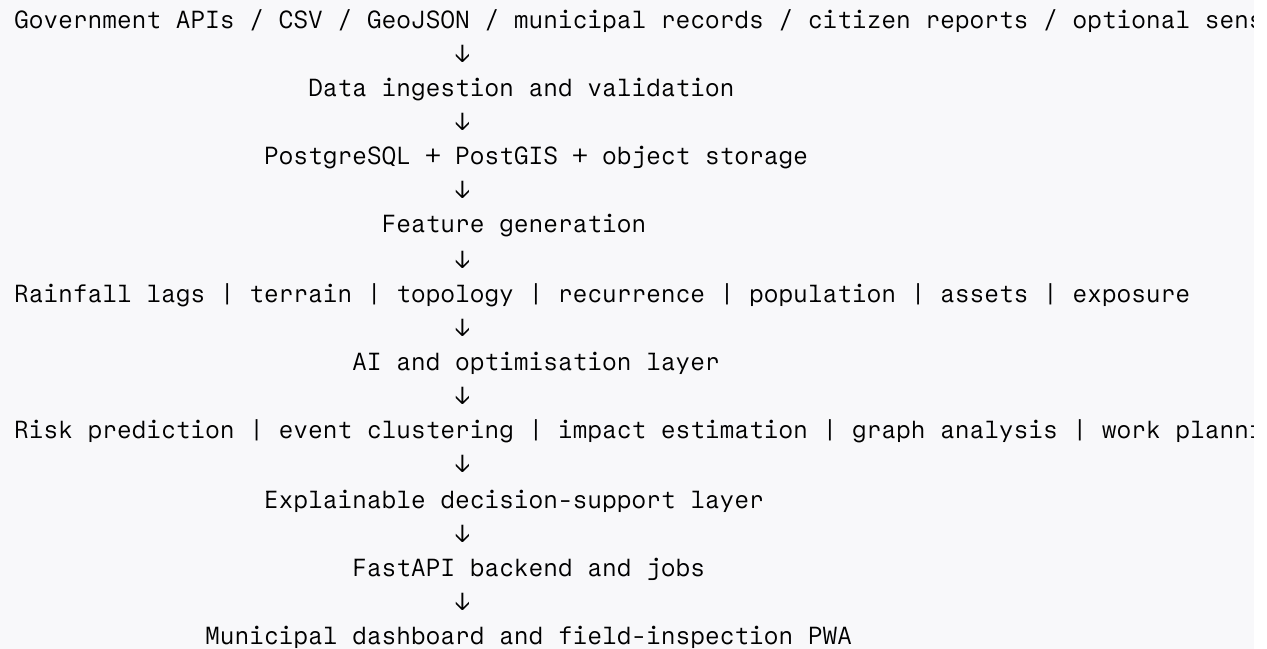
Example:

Blocked drain → Waterlogging → Road damage → Traffic disruption → Emergency response delay.

12. Closure Verification

Repair complete nu mark pannina piragu issue actually solve aachaa-nu future complaints, field photos, and recurrence base-la verify pannum.

12. Technical Architecture



13. Recommended Technology Stack

Frontend

- React or Next.js.
- MapLibre GL or Leaflet.
- Tailwind CSS.
- Time slider.
- Risk, exposure, recurrence, graph, and intervention map layers.

Backend

- Python.
- FastAPI.
- PostgreSQL.
- PostGIS.
- Redis and Celery or scheduled jobs.
- REST APIs.
- GeoJSON support.

Machine Learning

- LightGBM or XGBoost for risk prediction.
- Logistic regression and seasonal-naive forecasting as baselines.
- SHAP for model explanation.
- Spatial-temporal clustering for duplicate and recurrence detection.
- NetworkX or PostGIS graph operations.
- OR-Tools for resource optimisation.
- YOLO only for focused image analysis.
- Indic ASR and multilingual NLP if voice reporting is included.
- LLM/RAG only for SOP retrieval, structured extraction, and explanations.

Deployment

- Docker.
- Docker Compose.
- Managed PostgreSQL/PostGIS.
- Cloud deployment.
- Scheduled ingestion jobs.
- Model-version tracking.
- Prediction and recommendation audit logs.

14. AI/ML Components

Civic Issue Classification

Input: Citizen text, speech transcript, image, and location.

Output: Issue category, severity indicators, and confidence.

Method: Lightweight image model, rule-assisted NLP, and multilingual text encoder.

Metrics: Macro-F1, precision, recall, per-class recall, and confusion matrix.

Duplicate and Incident Clustering

Input: Location, timestamp, issue type, text, and image features.

Output: Unified civic incident clusters.

Method: Spatial-temporal clustering with text or image embeddings.

Metrics: Pairwise precision/recall and cluster purity.

Risk Forecasting

Input:

- Historical incident counts.
- Rainfall totals and anomalies.
- Rainfall lag values.
- Elevation and flow indicators.
- Drainage and road connectivity.
- Population exposure.
- Recurrence patterns.
- Seasonal features.
- Infrastructure proxies.

Output: 24-hour, 48-hour, and 72-hour risk probability or risk class.

Recommended model: LightGBM or XGBoost.

Metrics:

- PR-AUC.
- Recall@top-k risk areas.
- Brier score.
- Calibration error.
- Time-based validation.
- Spatial holdout validation.

Asset Failure Prediction

Input: Asset age, type, material, repair history, rainfall, environment, and spatial variables.

Output: Failure hazard or vulnerability score.

Recommended model: Survival analysis or XGBoost when actual asset labels are available.

Spatial Risk Analysis

Input: Geotagged incidents, road network, drainage network, elevation, and waterbodies.

Output: Hotspots, propagation paths, and risk corridors.

Methods: KDE, Moran's I, network analysis, flow accumulation, and graph propagation.

Impact Estimation

Input: Risk polygons/cells, population grids, roads, schools, hospitals, and public assets.

Output: Estimated exposed people and assets.

Method: Population-weighted spatial overlay.

Limitation: This is an estimated exposure value, not a verified disaster-impact count.

Work-Order Optimisation

Input: Predicted risk, population impact, intervention type, crew availability, skills, distance, time, cost, and ownership.

Output: Feasible inspection and maintenance schedule.

Method: OR-Tools constraint programming or mixed-integer optimisation.

Metrics:

- Risk reduction per budget unit.
- Population exposure reduced.
- Travel distance saved.
- Critical locations addressed.
- Schedule feasibility.

15. India-Specific Design Principles

CivicPulse AI generic US or European smart-city model illa. Indian urban conditions consider pannu design pannapadum.

Important principles:

- Limited municipal staff-ai consider pannum.
- Fragmented and incomplete data-la work pannum.
- Sensors mandatory nu assume pannaadhu.
- Monsoon preparedness focus pannum.
- Dense settlements and underreported areas include pannum.
- Population exposure and critical facilities consider pannum.
- Multilingual and low-literacy reporting support pannum.
- CSV and GeoJSON legacy data import support pannum.
- Low-bandwidth map and field workflow provide pannum.
- Department jurisdiction respect pannum.
- Human approval and explainability provide pannum.

- Unsupported certainty avoid pannum.
- Public data-la work aagum; official data add pannumbodhu improve aagum.

16. Data Sources

Rainfall and Weather

- Open Government Data rainfall catalogue.
- IMD public API resources.
- Central Water Commission rainfall resources.
- District and station-level rainfall.
- Historical rainfall normals and anomalies.

Geospatial Data

- OpenStreetMap roads, drains, waterways, buildings, schools, hospitals, and public facilities.
- Digital elevation and terrain layers.
- Sentinel-2 multispectral imagery.
- Sentinel-1 radar imagery.
- NRSC/NDEM flood hazard and inundation layers.

Population and Exposure

- Census 2011 population and density data.
- Gridded population data.
- Schools, hospitals, transport corridors, and public facilities.

Civic Incident Data

- Municipal complaint records where available.
- BBMP Fix My Street data.
- BBMP grievance data.
- OpenCity datasets.
- Clearly labelled synthetic or semi-synthetic incidents for prototype training and simulation.

17. MVP Scope

Hackathon-la realistically build panna vendiya MVP:

1. One Indian ward or small study area.
2. One focused hazard: drainage-related waterlogging.
3. 24-hour, 48-hour, and 72-hour forecast.
4. Public rainfall, OSM, population, and civic-event data.
5. Transparent synthetic-event generator if real labels are insufficient.
6. Baseline model and ML model comparison.
7. Grid-level risk prediction.
8. SHAP-based risk explanation.
9. Population and critical-facility exposure.
10. Drainage dependency graph.
11. Top-five preventive inspection recommendations.
12. Crew-constrained work plan.
13. Rainfall and intervention scenario simulator.
14. Public-data, observed-data, or simulation label.
15. Human-verification warning on every recommendation.

Build panna koodaadha features during hackathon:

- Full nationwide model.
- Live sensor network.
- Complete municipal integration.
- Full digital twin.
- All civic categories at once.
- Unsupported real-world failure claims.

18. Advanced Version

Future-la add panna mudiyura features:

- Water-pipe failure prediction.
- Pressure and flow sensors.
- Hydraulic simulation.
- Satellite-based flood validation.
- GNN-based infrastructure dependency learning.
- Multilingual voice reporting.
- Offline field-worker application.

- Automatic before-and-after repair verification.
- Cross-city transfer learning.
- Federated learning between ULBs.
- Live work-order integration.
- Digital-twin calibration.
- Causal intervention-effect estimation.

19. Two-Minute Judge Demo

1. Demonstration ward select pannunga.
2. Historical complaint map show pannunga.
3. Complaint map reactive and incomplete-nu explain pannunga.
4. 48-hour heavy-rainfall scenario activate pannunga.
5. New complaint varradhukku munnadi predicted risk map show pannunga.
6. Highest-risk location click pannunga.
7. Rainfall anomaly, recurrence, low elevation, drainage connectivity, population, and nearby school/hospital show pannunga.
8. Dependency graph open panni probable upstream drain/outlet identify pannunga.
9. Available crews five-lendhu two-aa reduce pannunga.
10. Resource optimiser run pannunga.
11. Preventive inspection order show pannunga.
12. Oru drain clear pannura scenario simulate pannunga.
13. Exposure reduction and residual risk show pannunga.
14. Final warning display pannunga: “AI recommendation; field verification required.”

20. Measurable Evaluation Metrics

Prediction Metrics

- PR-AUC.
- Recall@top-10 risk cells.
- Critical-risk precision.
- Brier score.
- Calibration error.
- Event-kku munnadi lead time.
- Spatial and temporal holdout performance.

Operational Metrics

- Scenario-la population exposure reduction.
- Budget unit-kku risk reduction.
- Crew travel distance saved.
- High-risk locations covered.
- Preparedness plan generate panna required time.
- Duplicate incidents merged.
- Evidence irukkura recommendations percentage.
- Human inspection required recommendations percentage.

Fairness and Reliability Metrics

- High-reporting and low-reporting areas coverage.
- Ward-wise performance.
- Missing-data sensitivity.
- Rainfall uncertainty sensitivity.
- Complaint data reduce pannumbodhu model performance.

21. Risks and Mitigation

Real Failure Labels Illaama Irukkum

Public incidents, manual labels, and transparent semi-synthetic simulation use pannunga. Possible-aana time-separated held-out events-la evaluate pannunga.

Data Leakage

Time-based split and spatial holdout use pannunga. Same location repeated incidents training and testing-la mix aaga koodaadhu.

Incomplete Drainage Data

OSM and inferred network-ai proxy-aa treat pannunga. Coverage confidence show pannunga. Officer infrastructure edit/verification option kudunga.

False Precision

Calibration, confidence bands, evidence cards, and inspect-before-action category use pannunga.

Weather Forecast Error

Forecast update aagumbodhu risk recalculate pannunga. Weather-data confidence show pannunga.

Old Population Data

Current census certainty nu sollaama estimated exposure nu display pannunga.

Complaint Bias

Population, reporting coverage, and independent geographic evidence use panni normalise pannunga.

Scope Too Broad

One hazard, one ward, and three decisions mattum focus pannunga:

- Enga risk high?
- Yaar affect aaga poraanga?
- Endha action first?

Government Adoption Barriers

CSV/GeoJSON import, human approval, explainability, offline inspection, and department ownership support pannunga.

Privacy

Personal information expose panna koodaadhu. Reports aggregate panni minimum necessary location precision mattum display pannunga.

Hallucinated Recommendations

Deterministic action templates and verified municipal SOPs use pannunga. LLM risk values or operational instructions invent panna allow panna koodaadhu.

Synthetic Data Trust Issue

Every synthetic layer label pannunga. Public-data, observed-data, and simulation modes separate-aa show pannunga.

22. Final Project Statement

CivicPulse AI enbadhu Indian urban local bodies-kkaaga design pannappatta explainable predictive civic infrastructure intelligence platform.

Adhoda focused implementation, CivicPulse Monsoon, rainfall, historical civic incidents, terrain, geospatial infrastructure relationships, population exposure, and recurrence patterns-

ai combine panni drainage and waterlogging risk-ai 24–72 hours munnadi predict pannum.

System probable upstream cause-ai identify pannum, affected population and critical facilities-ai estimate pannum, preventive inspections-ai prioritise pannum, limited municipal crews and budget-ai optimise pannum, and rainfall/intervention scenarios-ai simulate pannum.

CivicPulse AI oru normal complaint-management app illa.

Idhu municipal authorities-kku following questions answer panna help pannura preventive decision-support system:

- Next civic infrastructure problem enga varum?
- Eppo critical aagum?
- Yen andha location risk-la irukku?
- Yaar andha problem-naala affect aaga poraanga?
- Endha infrastructure responsible-aa irukkalam?
- Endha department action edukkanum?
- Endha action first edukkanum?
- Available crew and budget-la enna best plan?
- Intervention apram evlo risk remain aagum?

23. Final Pitch

CivicPulse Monsoon complaint varradhukku munnadi drainage-related civic failures enga nadakka pogudhu-nu predict pannum, yen risk irukku-nu explain pannum, yaar affect aaga poraanga-nu estimate pannum, and limited municipal crews and budget-kku best preventive inspection plan generate pannum.

24. Final Recommendation

Original broad concept-ai MODIFY pannunga.

CivicPulse Monsoon-aa focused, explainable, low-data, India-specific predictive system-aa build pannunga.

Main focus:

- Drainage and waterlogging risk prediction.
- Infrastructure dependency analysis.
- Population and critical-facility impact estimation.
- Preventive inspection prioritisation.
- Municipal crew and budget optimisation.
- Rainfall and intervention simulation.

Indha focused version technically credible-aa irukkum, measurable-aa irukkum, hackathon-la realistically build panna mudiyum, and normal civic complaint apps-lendhu clearly

differentiate aagum.